

Proteomics, metabolomics, and immunomics on microparticles derived from human atherosclerotic plaques.

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BACKGROUND: Microparticles (MPs) with procoagulant activity are present in human atherosclerosis, but no detailed information is available on their composition. METHODS AND RESULTS: To obtain insights into the role of MPs in atherogenesis, MP proteins were identified by tandem mass spectrometry, metabolite profiles were determined by high-resolution nuclear magnetic resonance spectroscopy, and antibody reactivity was assessed against combinatorial antigen libraries. Plaque MPs expressed surface antigens consistent with their leukocyte origin, including major histocompatibility complex classes I and II, and induced a dose-dependent stimulatory effect on T-cell proliferation. Notably, taurine, the most abundant free organic acid in human neutrophils, which scavenges myeloperoxidase-catalyzed free radicals, was highly enriched in plaque MPs. Moreover, fluorescent labeling of proteins on the MP surface suggested immunoglobulins to be trapped inside, which was confirmed by flow cytometry analysis on permeabilized and nonpermeabilized plaque MPs. Colabeling for CD14 and IgG established that more than 90% of the IgG containing MPs were CD14(+), indicating a macrophage origin. Screening against an antigen library revealed that the immunologic profiles of antibodies in MPs were similar to those found in plaques but differed profoundly from antibodies in plasma and unexpectedly, showed strong reactions with oligosaccharide antigens, in particular blood group antigen A. CONCLUSIONS: This study provides the first evidence that immunoglobulins are present within MPs derived from plaque macrophages, that the portfolio of plaque antibodies is different from circulating antibodies in plasma, and that anticarbohydrate antibodies are retained in human atherosclerotic lesions.